Nurullah Said Tegmen

180720823

$k = 8$

$L = 8$

$M = 2$

$N = 3$

Q1: $\varepsilon_{r_3} = 8$

breakdown strength $= 36 \frac{KV}{cm}$

$\theta_1 = 69$

$\theta_2 = 21^\circ$

$\theta_3 = 21^\circ$

$\theta_4 = 48^\circ$

a) $E_3 . \sin\alpha_4 = E_2 . \sin\alpha_3$

$36 . \underbrace{\sin(90-48)}_{0.67} = E_2 \underbrace{\sin(90-21)}_{0.93}$

$E_{2\,minimum} = 25.93 \frac{KV}{cm}$

$E_2 . \sin\alpha_2 = E_1 . \sin\alpha_1$

$25.93 . \underbrace{\sin(90-21)}_{0.93} = E_1 \underbrace{\sin(90-64)}_{0,44}$

$E_{1\,minimum} = 54.81 \frac{KV}{cm}$

b) $\frac{\tan\alpha_2}{\tan\alpha_3} = \frac{\varepsilon_2}{\varepsilon_3} \longrightarrow \frac{\varepsilon_{r_2}}{\varepsilon_{r_3}} = \frac{\tan(90-21)}{\tan(90-21)} \longrightarrow \varepsilon_{r_2} = \varepsilon_{r_3}$

$8 = 8$

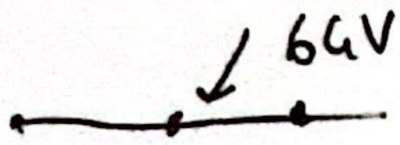
$\varepsilon_{r_2} = 8$

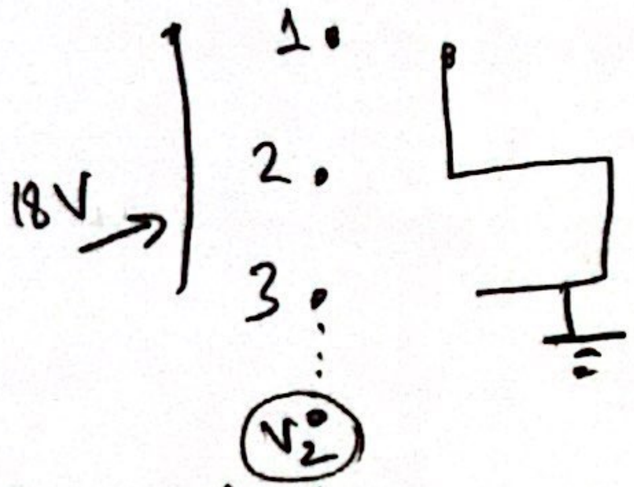
$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\varepsilon_1}{\varepsilon_2} \longrightarrow \frac{\varepsilon_{r_1}}{\varepsilon_{r_2}} = \frac{\tan(90-64)}{\tan(90-21} \longrightarrow \varepsilon_{r_1} = 8 \times 0.48$

$\varepsilon_{r_1} = 3.84$

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Nurullah Said Tuğman
~~Said~~ 150720913 $\frac{L}{100}$ +0.50

2) 



K = 8
L = 8
M = 2
N = 3

for node 1 :

$$V_1 = \frac{1}{4}(18 + 64 + 0 + V_2) \rightarrow V_1 = 20.5 + \frac{V_2}{4}$$

$$4V_1 = 82 + V_2 \therefore$$

for node 2:

$$V_2 = \frac{1}{4}(18 + V_1 + 0 + V_3) \rightarrow V_2 = 4.5 + V_1 + V_3 \therefore$$

for node 3:

$$V_3 = \frac{1}{4}(18 + 0 + 2V_2) \rightarrow V_3 = 4.5 + \frac{V_2}{2}$$

$$2V_3 = 9 + V_2 \therefore$$

$$V_2 = 4.5 + \overbrace{20.5 + \frac{V_2}{4}}^{V_1} + \overbrace{4.5 + \frac{V_2}{2}}^{V_3} \rightarrow V_2 = 29.5 + \frac{3V_2}{4}$$

$$4V_2 = 118 + 3V_2$$

$V_1 = 50\,kV$

$V_3 = 63.5\,kV$

$V_2 = 118\,kV$

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150720823

K = 8
L = 8
M = 2
N = 3

3) diameter → 120 mm = 12 cm → r = 6 cm

correction factor : $\underbrace{\frac{L}{100}}_{0.98} + 0.9 = 0.98$

$$E_{max} = k\frac{U/2}{r\ln\left(\frac{r+d/2}{r}\right)} \Rightarrow 30\frac{kV}{cm} = 0.98 \cdot \frac{\frac{390}{2}}{6\ln\left(\frac{6+d/2}{6}\right)} = \frac{0.98 \times 190}{6} \cdot \frac{1}{\ln(}$$

$$\ln\left(\frac{6+d/2}{6}\right) = 1.03 \qquad \frac{6+d/2}{6} = 2.8 \qquad \frac{d}{2} + 6 = 16.8$$

$$\boxed{d = 21.6\,cm} \qquad \frac{d}{2} = 10.8$$

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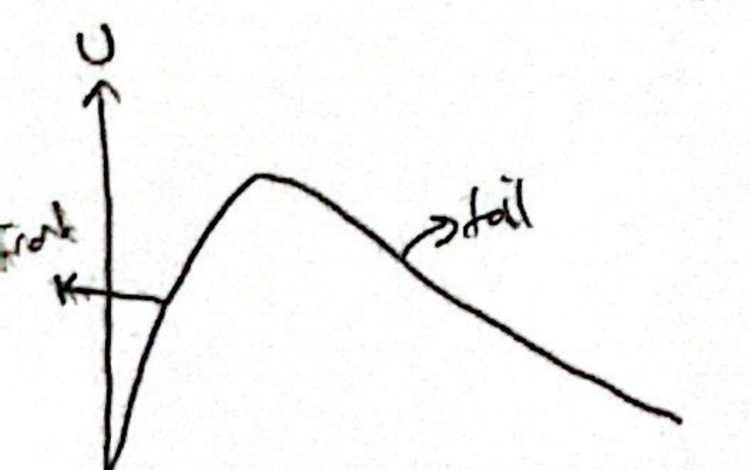
4) a) To analyzing HV test results, some of them work with machine learning and deep learning. We use silicon in industry for insulation.

b) ① → High voltage temperature humidity bias test

② → Impulse voltage test

③ → Partial discharge test

Impulse voltage test:



this test is using for generating a switching impulse, it aims to prevent from natural lightning. So before the natural lightning, this test check the system whether it is durable or not.

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